

AI for Financial Practitioners

Lecture 1 · Part I

From Statistical Models to Generative Intelligence

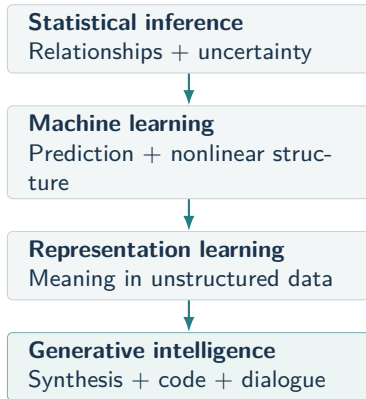
The executive shift: from model user to architect of intelligent work

Finance did not leave statistics behind

CONTINUITY, NOT REPLACEMENT

Modern AI expands the operations we can perform on representations of an uncertain world.

- A balance sheet, yield curve, regression and credit score are all **representations**.
- AI adds richer representations of **language, documents, images and sequences**.
- Generative systems add **synthesis, transformation and interaction** through natural language.



Four capabilities — four different financial questions

KNOW WHAT CLAIM THE SYSTEM IS MAKING

01 · ESTIMATE

Question

What relationship
is present?

Finance

Beta · elasticity
· sensitivity

02 · PREDICT

Question

What is likely to
happen?

Finance

Default · fraud ·
return · flow

03 · REPRESENT

Question

What information
is relevant?

Finance

Filings · calls ·
contracts · news

04 · GENERATE

Question

How can evidence
be transformed?

Finance

Memo · code
· scenarios ·
dialogue

Prediction is not decision

THE MOST IMPORTANT CATEGORY DISTINCTION

7%

Probability of default

A forecast describes a possible state of the world.

The number cannot tell the portfolio manager what to do.

Expected recovery + spread

Liquidity + transaction cost

Portfolio correlations + exposure

Mandate + risk limits + liabilities

Alternatives + cost of being wrong

Legitimate authority to act

Generative AI changes the interface — not the laws of finance

NATURAL LANGUAGE BECOMES AN INTERFACE TO ANALYTICAL WORK

Business question · Why did free cash flow deteriorate?



Retrieve · filings, footnotes, calls, prior periods



Calculate · reconcile accounting definitions



Synthesize · competing explanations + sensitivities



Challenge · “show the bridge” · “argue against our view”

What becomes scarce?

- Domain judgment
- Precise task specification
- Definition discipline
- Ability to test the answer

Prototype ≠ production

A successful conversation proves possibility—not repeatability, control or accountability.

A dependable AI system is an architecture, not a model

ASSIGN EACH RESPONSIBILITY TO THE CAPABILITY BEST SUITED TO CARRY IT

CONTEXT · entity · mandate · period · user · workflow state

RETRIEVAL · current, authorized, inspectable evidence

MEMORY · approved state · provenance · timestamps · revalidation

DETERMINISTIC TOOLS · arithmetic · pricing · database queries · code execution

GENERATIVE MODEL · interpretation · synthesis · interaction · orchestration

HUMAN AUTHORITY · objectives · contested judgment · consequential action

Do not let claims inherit authority from one another

A PRACTICAL TAXONOMY FOR EVERY AI-PRODUCED FINANCIAL NOTE

Claim	Example	What must support it
Descriptive	Net debt increased	Entity · date · unit · source · definition
Inferential	Acquisition drove leverage	Interpretive / causal argument + alternatives
Predictive	Breach risk is rising	Horizon · calibration · relevant population
Generative	Two-page executive memo	Faithfulness · completeness · usefulness
Prescriptive	Reduce exposure	Objectives · constraints · alternatives · authority

Evaluate the workflow — not the demo

FROM “IMPRESSIVE ONCE” TO “DEPENDABLE IN PRODUCTION”

Define a bounded responsibility

“Identify material changes in liquidity risk from newly published issuer documents and produce a cited draft for analyst review within 30 minutes.”

Compare against the real baseline

Measure **useful completion**: total quality and cost of getting the work to an acceptable state.

Evidence · right source, right version

Numbers · reconcile to controlled definitions

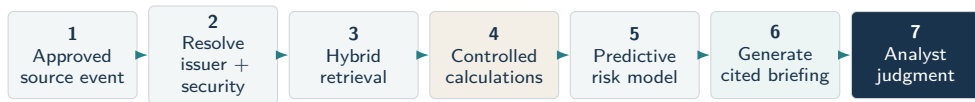
Usefulness · finds what changes the decision

Operations · latency, cost, reviewer burden

Control behavior · abstain, escalate, record

Worked case: an AI credit-monitoring service

BOUND THE RESPONSIBILITY: DETECT DETERIORATION AND PREPARE AN ANALYST-READY BRIEFING



The system may

- Detect new evidence
- Calculate leverage / coverage / maturities
- Surface competing explanations
- Prioritize cases and scenarios

The system does not infer

- The portfolio's complete objective
- The acceptable risk budget
- Relative-value alternatives
- Authority to trade

The executive playbook: move from AI user to AI architect

FIVE QUESTIONS BEFORE CHOOSING A MODEL OR VENDOR

01 · What **responsibility** are we improving?

02 · What claims require **evidence, calculation or prediction**?

03 · Which capability should carry **each responsibility**?

04 · Where is the **action boundary** and who has authority?

05 · What evidence would prove the workflow is **better than today's**?

The shift

From
“What can this model do?”

To
“**How should intelligence be assembled for this task?**”

Capability → architecture → process